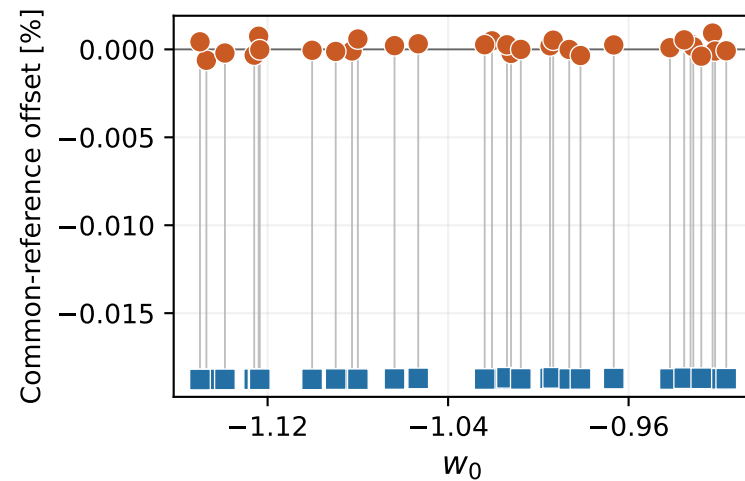


Paired quadrature checks: shared_narrow_peak

Common reference: the same spectrum evaluated with CCL kernels and GSL CQUAD

Split circles: left = CCL QAG; right = CoCoA. Diamonds = deliberate search/control.

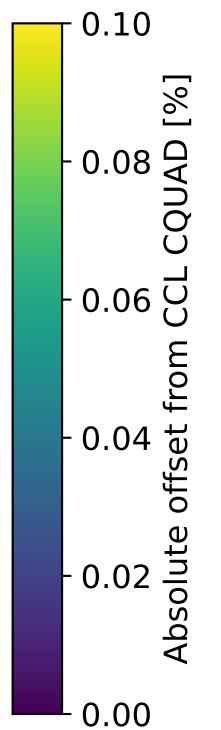
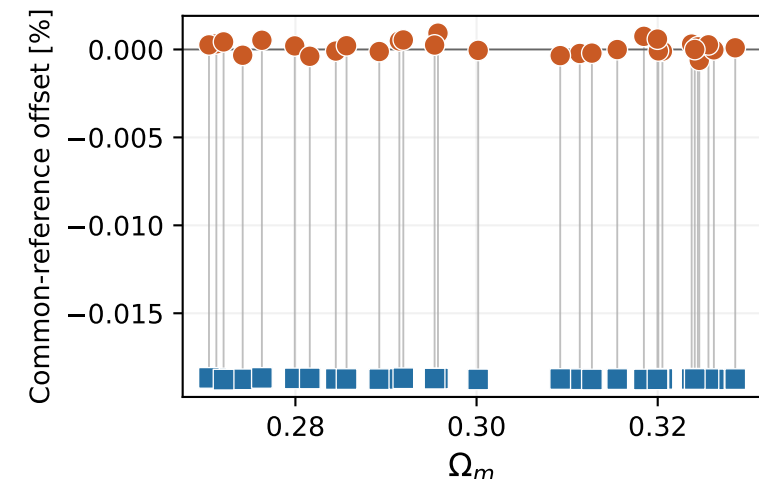
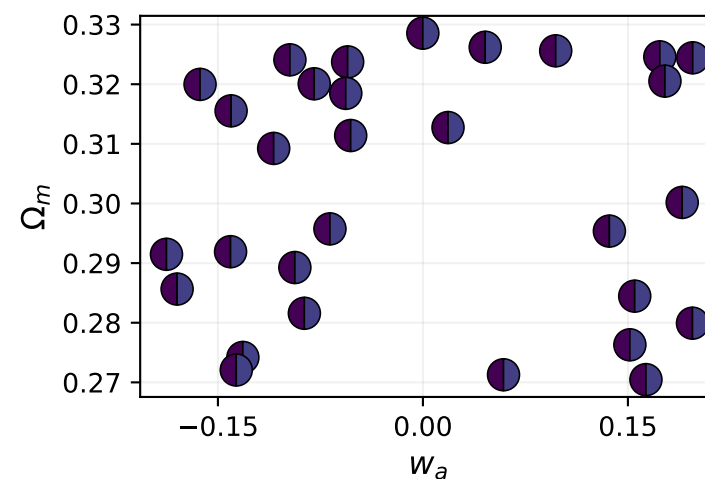
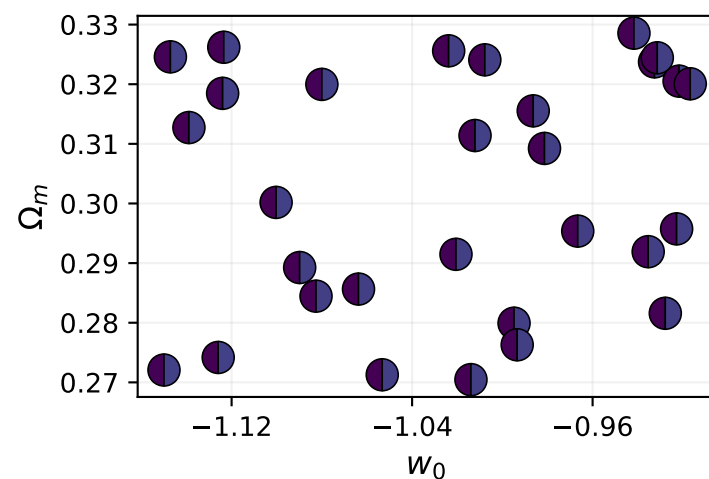
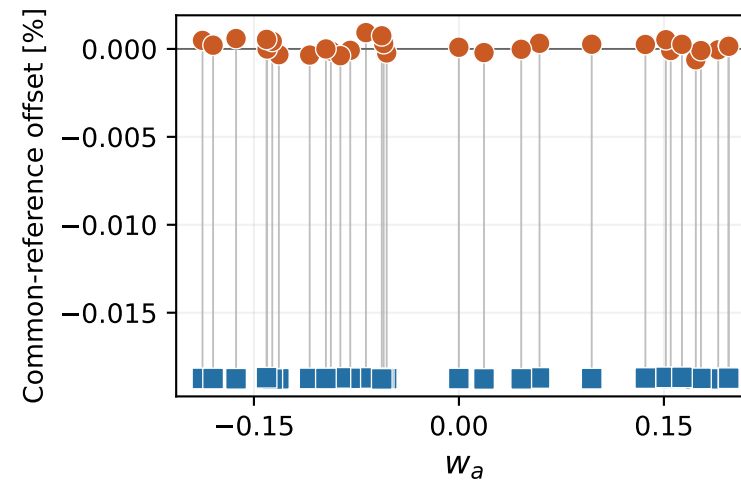
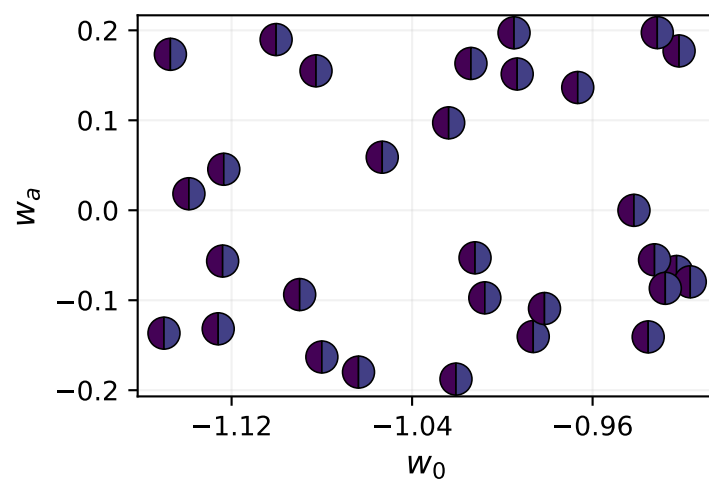
● CCL QAG ■ CoCoA 96-node spectra



Different questions, different references

Code	Own quadrature	Common reference
CCL QAG	0.00092629%	0.00092629%
CoCoA	0.005268%	0.018774%

Largest absolute deviations among displayed cases.
Own quadrature: QAG vs CQUAD; CoCoA 96 vs 1024.
Common reference: both codes vs CCL CQUAD.
A cross-code offset is not itself quadrature error.



Red outline: either code exceeds 0.1% own-reference quadrature error (0 displayed cases).

30 uniform cases available; 0 targeted/control cases. Diagnostic coordinates, not posterior samples.

CoCoA: actual covariance-module all-pairs spectra. No covariance matrix; no ordinary data-vector cross-bin test.

Grey crosses: unresolved or not yet paired. Targeted points are excluded from frequency estimates. No universal immunity claim.